



Childhood Tracheobronchial Tumors Treatment (PDQ®)–Health Professional Version

Types of Childhood Tracheobronchial Tumors

Primary lung tumors are rare in children and histologically quite diverse.[1] A review of primary malignant epithelial lung tumors using the National Cancer Database found that the most common primary malignant pediatric lung neoplasms were carcinoid tumors (63%), followed by mucoepidermoid carcinoma of the lung (18%).[2]

Tracheobronchial tumors are a heterogeneous group of primary endobronchial lesions. Although adenoma implies a benign process, all varieties of tracheobronchial tumors occasionally display malignant behavior. Most primary lung tumors are malignant. In a review of 383 primary pulmonary neoplasms in children, 76% were malignant and 24% were benign. [3]

The following histological types have been identified (see [Figure 1](#)):[4-10]

- **Carcinoid tumors (neuroendocrine tumors of the bronchus).** Carcinoid tumors account for 80% to 85% of all tracheobronchial tumors in children.[4-8]; [11][[Level of evidence C1](#)] They are the most common tracheobronchial tumor.
- **Mucoepidermoid carcinomas.** Slow-growing vascular polypoid masses of the airway that are the second most common (10%) pediatric tracheobronchial tumor.
- **Inflammatory myofibroblastic tumors.** These low-grade tumors account for 1% of pediatric tracheobronchial tumors, are commonly located in the upper trachea, and rarely metastasize. Intrabronchial (pulmonary) inflammatory myofibroblastic tumors are incredibly rare compared with other sites of disease. For more information, see [Childhood Pulmonary Inflammatory Myofibroblastic Tumors Treatment](#).
- **Rhabdomyosarcomas.**
- **Granular cell tumors.** Malignant transformation of these tumors has not been documented in pediatric patients.

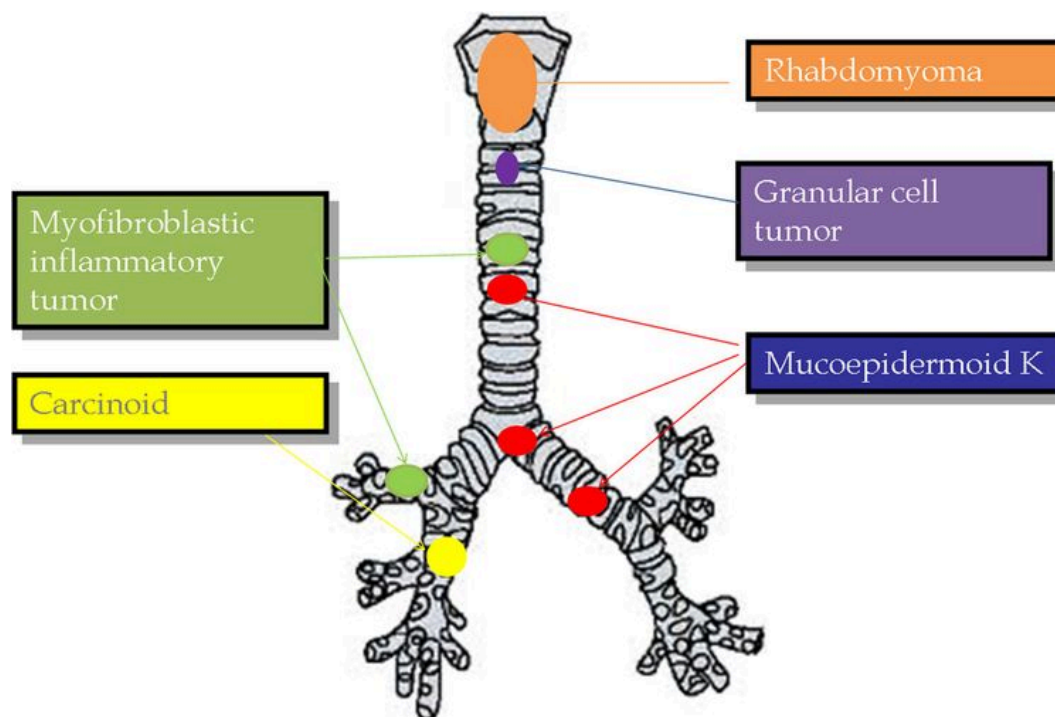


Figure 1. The most representative primary tracheobronchial tumors are described with their more frequent location. Reprinted from [Seminars in Pediatric Surgery](#), Volume 25, Issue 3, Patricio Varela, Luca Pio, Michele Torre, Primary tracheobronchial tumors in children, Pages 150–155, Copyright (2016), with permission from Elsevier.

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Clinical Presentation

When epithelial cancers of the lung occur, they tend to be of advanced stage. Prognosis is dependent on both histology and stage.[\[1\]](#)

The presenting symptoms of tracheobronchial tumors are usually caused by an incomplete tracheobronchial obstruction and include the following:

- Cough.
- Recurrent pneumonitis.
- Hemoptysis.

Because of difficulties in diagnosis, symptoms are frequently present for months. Occasionally, children with wheezing are treated for asthma, and diagnosis can be delayed for as long as 4 to 5 years.[\[2\]](#)

Metastatic lesions are reported in approximately 6% of patients with carcinoid tumors, and recurrences are reported in 2% of cases. Atypical carcinoid tumors are rare but more aggressive, and 50% of these patients present with metastatic disease at diagnosis.[\[1,3\]](#) One study reported a single child with a carcinoid tumor and metastatic disease who developed the classic carcinoid syndrome.[\[4\]](#) Octreotide nuclear scans may demonstrate uptake of radioactivity by the tumor or lymph nodes, suggesting metastatic spread.

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Diagnostic Evaluation

The management of tracheobronchial tumors is somewhat controversial because tracheobronchial tumors are usually visible endoscopically.

Biopsy of these lesions may be hazardous because of the risk of hemorrhage. New endoscopic techniques have allowed biopsy to be performed more safely.[\[1,2\]](#) However, endoscopic resection is not recommended except in highly selected cases.[\[2-4\]](#)

Bronchography or computed tomography may be helpful to determine the degree of bronchiectasis distal to the obstruction since the degree of pulmonary destruction may influence surgical therapy.[\[5\]](#)

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Prognosis

With the exception of rhabdomyosarcomas, tracheobronchial tumors of all histological types are associated with an excellent prognosis after surgical resection in children, even in the presence of local invasion.[1,2]; [3][[Level of evidence B4](#)]

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Treatment of Childhood Tracheobronchial Tumors

The treatment of choice is conservative pulmonary resection, including sleeve segmental resection (when feasible), with the removal of the involved lymphatics.[1,2]; [3][[Level of evidence B4](#)]

Chemotherapy and radiation therapy are not indicated for tracheobronchial tumors, unless evidence of metastasis is documented or the tumor is a rhabdomyosarcoma.

Treatment options for tracheobronchial tumors, according to histological type, are as follows:

Carcinoid Tumors (Neuroendocrine Tumors of the Bronchus)

The treatment of choice is surgical resection with lymph node sampling. The overall survival rate is higher than 90% for these patients.[4-6] One study reported no deaths among patients who did not have lymph node involvement, and the 10-year survival rate was 89% for patients with one or more involved lymph nodes.[6]

Mucoepidermoid Carcinomas

The recommended treatment is open surgical resection and lymph node sampling. Endoscopic resection is not recommended.[7,8]

Inflammatory Myofibroblastic Tumors

The treatment of choice is surgery. However, if the tumor has an *ALK* variant, treatment with crizotinib may be effective.[\[8-11\]](#)

Rhabdomyosarcomas

Surgery is not indicated. These tumors are very responsive to chemotherapy and radiation therapy, even with lymph node metastasis.[\[8\]](#) For more information, see [Childhood Rhabdomyosarcoma Treatment](#).

Granular Cell Tumors

Surgical resection is based on morbidity risk.[\[8,12,13\]](#)

For information about other neuroendocrine carcinoid tumors, see [Pediatric Gastrointestinal Neuroendocrine Tumors Treatment](#).

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Treatment Options Under Clinical Evaluation for Childhood Tracheobronchial Tumors

Information about National Cancer Institute (NCI)–supported clinical trials can be found on the [NCI website](#). For information about clinical trials sponsored by other organizations, see the [ClinicalTrials.gov website](#).

Special Considerations for the Treatment of Children With Cancer

Cancer in children and adolescents is rare, although the overall incidence has slowly increased since 1975.[1] Children and adolescents with cancer should be referred to medical centers that have a multidisciplinary team of cancer specialists with experience treating the cancers that occur during childhood and adolescence. This multidisciplinary team approach incorporates the skills of the following pediatric specialists and others to ensure that children receive treatment, supportive care, and rehabilitation to achieve optimal survival and quality of life:

- Primary care physicians.
- Pediatric surgeons.
- Pathologists.
- Pediatric radiation oncologists.
- Pediatric medical oncologists and hematologists.
- Ophthalmologists.

- Rehabilitation specialists.
- Pediatric oncology nurses.
- Social workers.
- Child-life professionals.
- Psychologists.
- Nutritionists.

For specific information about supportive care for children and adolescents with cancer, see the summaries on [Supportive and Palliative Care](#).

The American Academy of Pediatrics has outlined guidelines for pediatric cancer centers and their role in the treatment of children and adolescents with cancer.[2] At these centers, clinical trials are available for most types of cancer that occur in children and adolescents, and the opportunity to participate is offered to most patients and their families. Clinical trials for children and adolescents diagnosed with cancer are generally designed to compare potentially better therapy with current standard therapy. Other types of clinical trials test novel therapies when there is no standard therapy for a cancer diagnosis. Most of the progress in identifying curative therapies for childhood cancers has been achieved through clinical trials. Information about ongoing clinical trials is available from the [NCI website](#).

Dramatic improvements in survival have been achieved for children and adolescents with cancer. Between 1975 and 2020, childhood cancer mortality decreased by more than 50%.[3-5] Childhood and adolescent cancer survivors require close monitoring because side effects of cancer therapy may persist or develop months or years after treatment. For information about the incidence, type, and monitoring of late effects in childhood and adolescent cancer survivors, see [Late Effects of Treatment for Childhood Cancer](#).

Childhood cancer is a rare disease, with about 15,000 cases diagnosed annually in the United States in individuals younger than 20 years.[6] The U.S. [Rare Diseases Act of 2002](#) defines a rare disease as one that affects populations smaller than 200,000 people in the United States. Therefore, all pediatric cancers are considered rare.

The designation of a rare tumor is not uniform among pediatric and adult groups. In adults, rare cancers are defined as those with an annual incidence of fewer than six cases per 100,000 people. They account for up to 24% of all cancers diagnosed in the European Union and about 20% of all cancers diagnosed in the United States.[7,8] In children and adolescents, the designation of a rare tumor is not uniform among international groups, as follows:

- A consensus effort between the European Union Joint Action on Rare Cancers and the European Cooperative Study Group for Rare Pediatric Cancers estimated that 11% of all

cancers in patients younger than 20 years could be categorized as very rare. This consensus group defined very rare cancers as those with annual incidences of fewer than two cases per 1 million people. However, three additional histologies (thyroid carcinoma, melanoma, and testicular cancer) with incidences of more than two cases per 1 million people were also included in the very rare group due to a lack of knowledge and expertise in the management of these tumors.[9]

- The Children's Oncology Group defines rare pediatric cancers as those listed in the International Classification of Childhood Cancer subgroup XI, which includes thyroid cancers, melanomas and nonmelanoma skin cancers, and multiple types of carcinomas (e.g., adrenocortical carcinomas, nasopharyngeal carcinomas, and most adult-type carcinomas such as breast cancers and colorectal cancers).[10] These diagnoses account for about 5% of the cancers diagnosed in children aged 0 to 14 years and about 27% of the cancers diagnosed in adolescents aged 15 to 19 years.[4]

Most cancers in subgroup XI are either melanomas or thyroid cancers, with other cancer types accounting for only 2% of the cancers diagnosed in children aged 0 to 14 years and 9.3% of the cancers diagnosed in adolescents aged 15 to 19 years.

These rare cancers are extremely challenging to study because of the relatively few patients with any individual diagnosis, the predominance of rare cancers in the adolescent population, and the small number of clinical trials for adolescents with rare cancers.

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